Full Scene Rendering and Cinematic

Advanced Rendering

When it came to the decisions behind what new technologies and what rendering features to implement, it was a bit of a rollercoaster.

For starters, many options such as Nanite, Substrate, and TSR, were difficult to justify using, as I could not tell the difference when they were enabled versus disabled, which I suppose is part of the point but I'll press on. I kept Nanite enabled as from an optimization standpoint, it sounds quite good without any noticeable downside, and I hoped it would help my colosseum mesh which has an incredibly high polycount that could affect performance. I didn't make much use of Substrate, as my scene didn't require any complex layering or stacking, and wasn't necessarily intended to look truly realistic. With TSR, I could not tell the difference between it or the other anti-aliasing methods. In the end, I did leave it enabled as I had previously encountered some blurriness in my scene (fixed with post-processing) and though I had fixed it, decided against potentially causing it to occur again.

Others such as Volumetric Fog and Water Systems I felt unnecessary, as I already had created shaders for these that were more performant and fit the visuals I wanted. I did however use the Volumetric Clouds for the sky, as I felt they helped tie the scenic design of the sky arena together. I also used the new Niagara System for my particle effects as that seems to be industry standard now for Unreal.

In terms of the rendering method, I stuck with deferred, as it fit my project a bit better. For starters, I did not really make use of any transparency, meaning I would not benefit in that regard for using forward rendering. As well, despite the bit more performance cost, it works well with systems like Lumen, which I did end up using.

Finally, for the lighting, as mentioned, I ended up using Lumen. Originally, I wanted to use more traditional methods and bake most of my lighting since I had primarily static lights. Though I tried this method, there were issues with the models that prevented me from properly baking to their lightmaps. With that, I decided to turn back to Lumen and try my best to properly optimize the light complexity throughout the scene. I did not look into ulterior ray tracing methods, as they are typically even more costly than Lumen, and I already felt that to be excessive. On top of that, since my scene was quite open, the added light bounce calculations I felt wouldn't amount to much, and that it would look quite similar to just using Lumen.

Optimizations

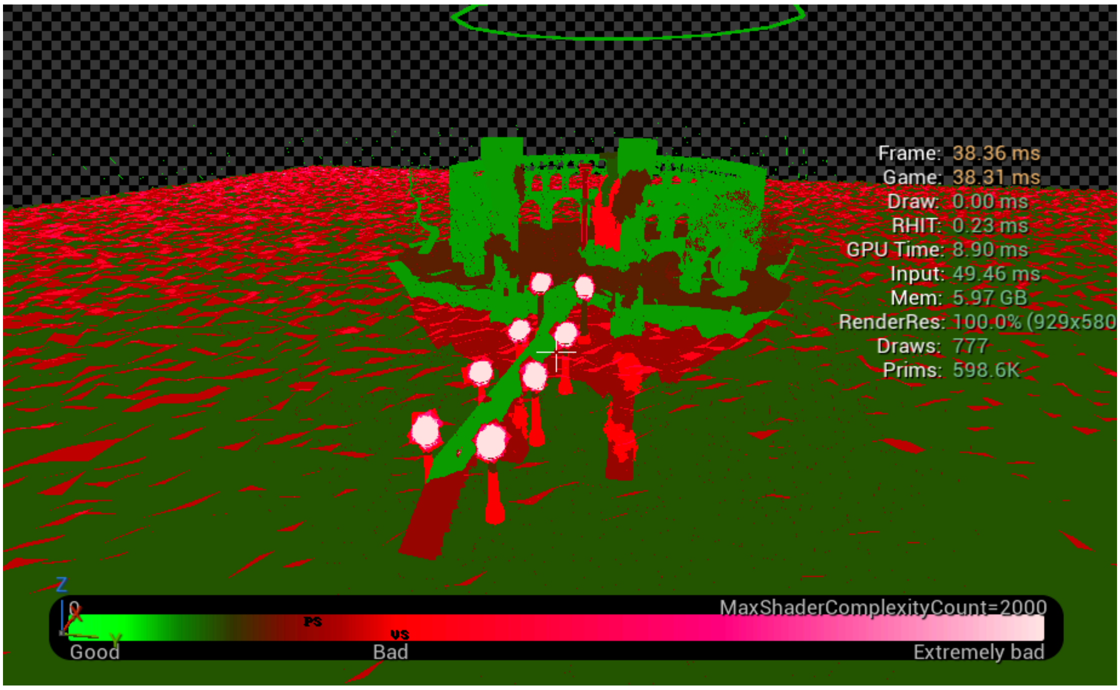
Shaders & Particles

Using the Shader Complexity view, I was able to pinpoint the shaders and particles that were most expensive.

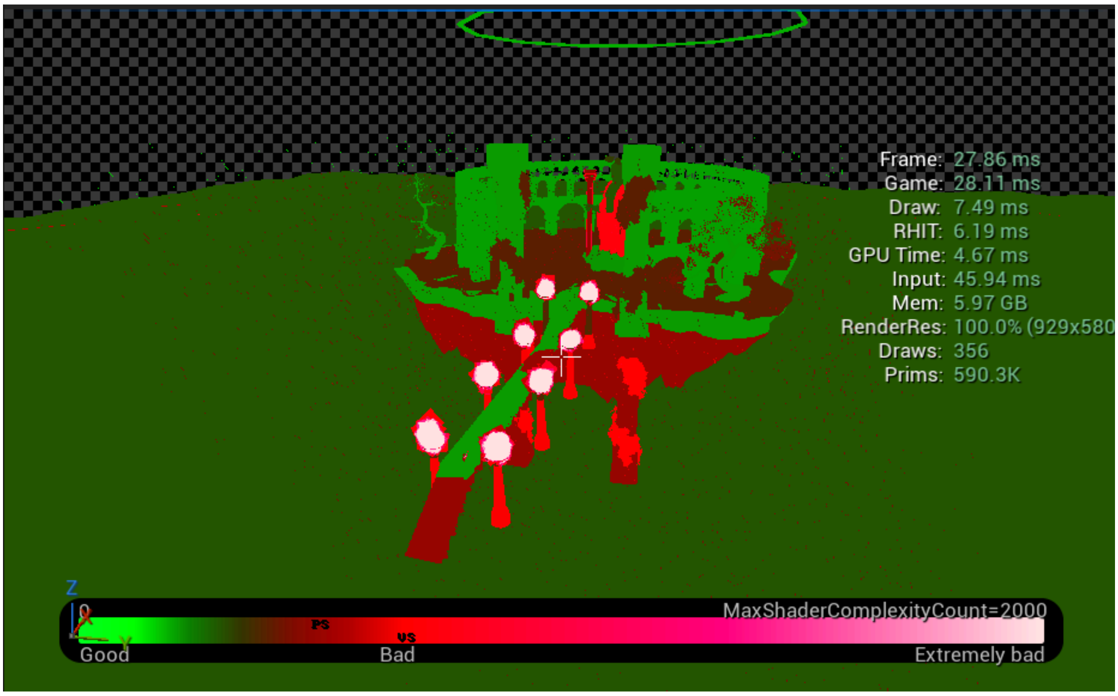
For the Shaders, this ended up primarily being the cloud sea, and any animated shader on objects that had excessive and overlapping polys. For the latter, I simply switched out the materials for ones that were static, immediately improving their performance cost. As for the former, the reason it seemed to be so heavy was due to the fact that it was somewhat transparent and rendered even when overlapping itself. Though this was impossible to completely fix with my knowledge of transparency, I did reduce the overall wavecount and their height to avoid having excessive overlaps. The same applied to the transparent and vertex-paint animated shaders on the boss. Due to the size of my scene and the fact that there is only one of each of the aforementioned elements, I believe the overall effect on performance is minimal (which was confirmed by using the stat profilers).

As for the Particles, this ended up being the fire, as it has many overlapping particles. I tried to optimize it by having them run on the GPUCompute Sim with a fixed bounds and by disabling all but the single fire particle (disabling smoke, debris, heat distortion) to some success, but ultimately they are still quite unoptimized. Without many other options, I decided to just keep the amount of them throughout the scene low, using only 6 on the exterior on the pathway outside the boss arena, which would allow them to be disabled during the boss fight in theory.

Before



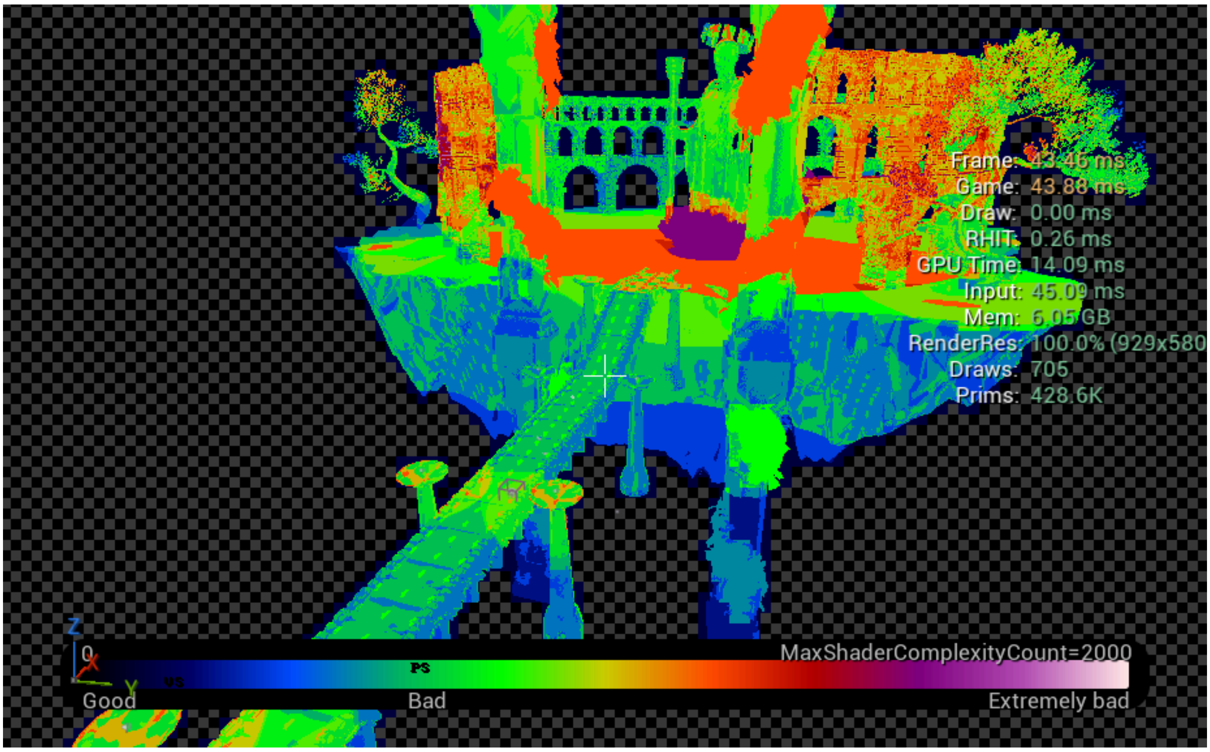
After



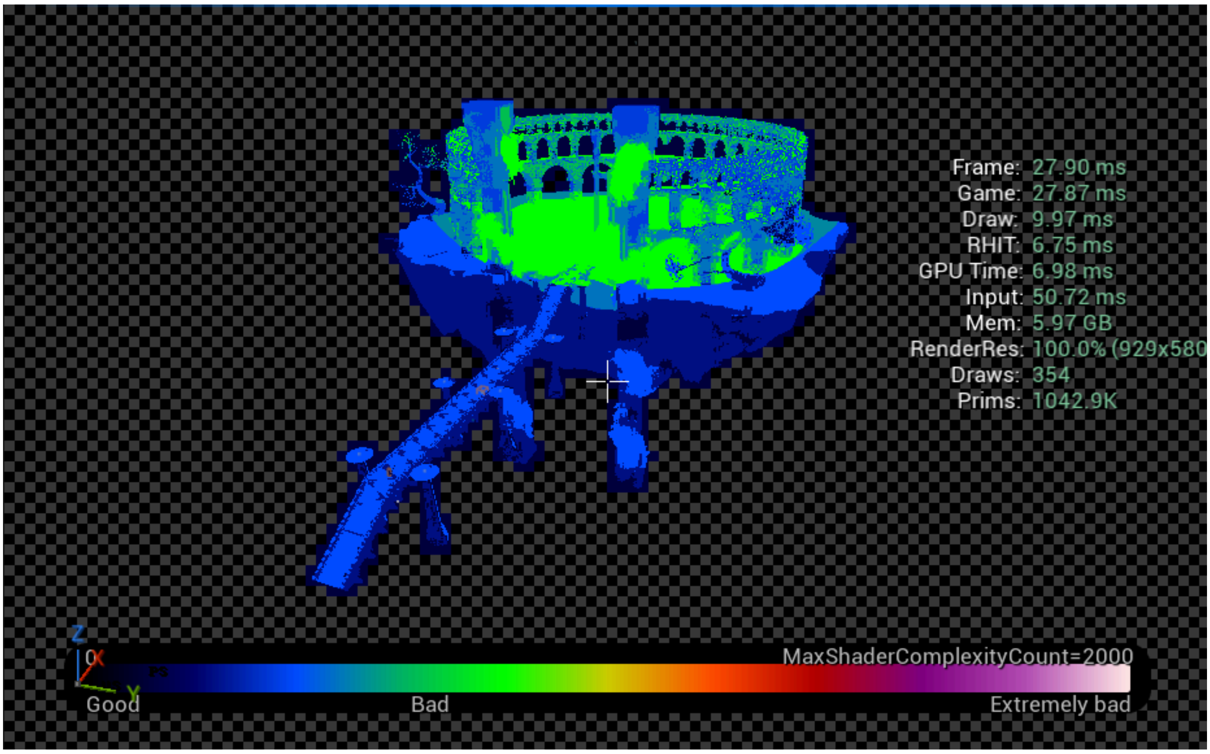
Lighting

Using the Light Complexity view, I was able to pinpoint areas of concern with my lighting. When it came to my original plan, I had wanted to use baking to drastically lower the overall cost of lighting throughout my scene. Unfortunately, I could not, so instead, I simply adjusted my lighting throughout my scene to reduce any unnecessary overlaps. This involved re-positioning or removing the light, or adjusting its attenuation, based on its importance and its use. For example, the lights within the fire pits used to be from the particle effect itself, but I found that a single point light accomplished the same for a bit better cost.

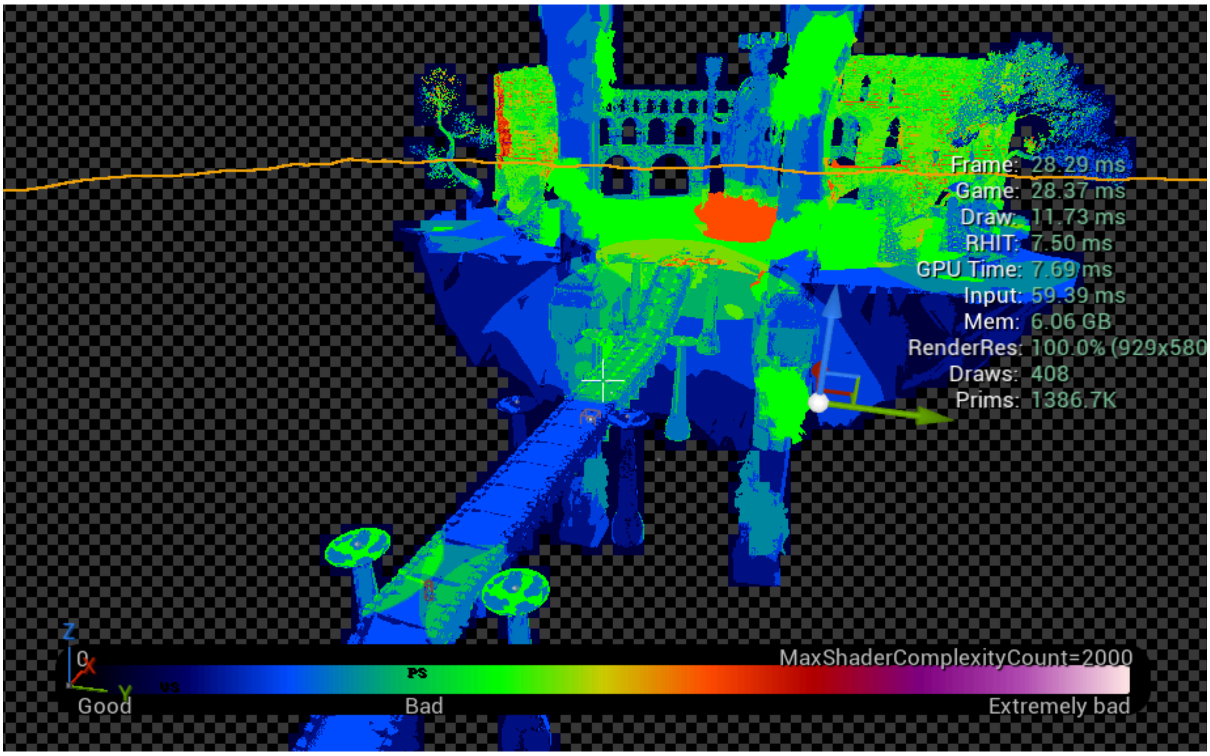
Before

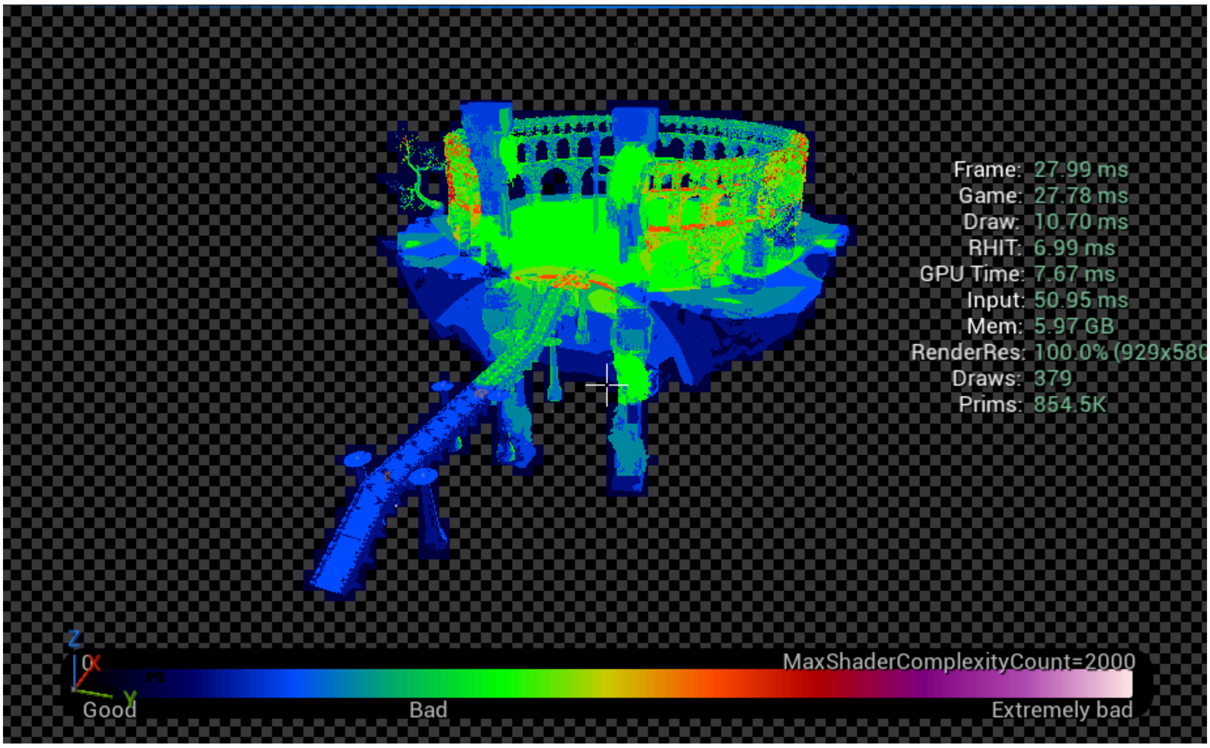


Planned (Baked Lighting)



After (Actual Scene Currently)





Scene Planning

Theming & Visual Design

For this assignment, I continued to use my original moodboard image as my primary reference. I didn't want to have too many influences going into this project, as I wanted to see what kind of scene I could come up with myself based on the initial "prompt" image.

When it came to the aesthetics of the environment, I wanted it to reflect the same feeling I got looking at the reference: a sense of awe. With that in mind, I wanted this primarily expressed through the size differences, framing, and colors.

Sizing was relatively simple; I kept the player small in scale while the environment and boss were scaled up. This, along with some framing during the opening cinematic to further accentuate the size difference, helped make the player seem puny and helpless in comparison to the entity and battlegrounds ahead.

With the color, I have always had a preference for the cold temperature ones, so I opted to make my scene primarily represented through them. I also felt that the cooler colors aided greatly to sell the time of day, which was night. Though the scene was primarily composed of cool colors, I did use some red and purples throughout to contrast the heavy use of blue and desaturation. These pops of additional color also accentuated important elements of the scene, which is why they were included on the two large entrance pillars and the boss's robes. Also, despite the darker lighting I was going for, I didn't want the scene to be too dim overall, and decided to use post-processing effects to balance it out. While keeping the scene dark, I boosted the colors to have the scene feel more vibrant. Along with that, I also strengthened the details as the scene looked a bit fuzzy, did some adjustments to the contrast and gamma, and added some bloom and vignette for a little bonus, as I felt it helped tie the scene's theming together a bit more (as one boosted the bright ethereal glows and the other added some realistic-ness to the dark of night).

Cinematics

When approaching how to do the cinematics, I primarily based it off trying to further push the intentions of the theming and previous boss encounters I found memorable.

Intro:
To push the theming, I knew I wanted to include the shot with the player and boss on opposite thirds to further draw to the immense difference in scale. As I knew my last shot, I worked backwards from there, and decided I'd have two other shots to show the character as he ran up to face his worthy foe.

The player runs up out of the sea of clouds, unaware of this new environment (wide shot - moving outward). He charges in headstrong with sword in hand, determination clear on his face, as he passes by the pillars of blue fire (wide shot to close-up - dolly sideways - high focal to have the foreground and background almost flat). As he approaches the arena, he takes a minute to take in the sight, readying himself for the encounter (wide shot - moving outward while rotating around the player - low focal to have the distance between them exaggerated).

Gameplay:
Used to walk around the scene and showcase the different visual elements throughout.

Outro:
Again, to align with the theming, I wanted the scale difference to be obvious once again, as I show the small player within the massive collapsing arena.

After coming out victorious, the island begins to shake (close-up cut to wide shot - stationary - player cut to pillar - high to low focal). With the death of its master, the arena crumbles and the island begins to fall from the sky (wide shot - dolly around front of island). The player quickly makes his escape, as the island sinks below the clouds (close-up to wide shot - moving backward - normal to high focal to show the environment completely).